

Ethical Dilemmas Facing Nephrology: Past, Present, and Future

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CHAPTER OUTLINE

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KEY POINTS

- The ethical issues facing nephrology played a fundamental role in the early development of the field of bioethics. Because the life-sustaining therapies of dialysis and kidney transplantation predated many parallel therapies in other fields, the ethical issues that inevitably accompany these highly specialized medical treatments were first addressed by nephrologists.
- The criteria for dialysis patient selection used by the Seattle Artificial Kidney Center Admissions and Policy Committee in the 1960s were widely criticized. It was deemed unacceptable to make life and death decisions based on a narrow definition of social worth.
- The US Medicare End-Stage Renal Disease (ESRD) program quickly grew beyond initial projected number of covered patients in part because the dialysis community began treating older patients with more comorbidities. These included patients with other life-threatening illnesses such as end-stage cardiovascular disease, debilitating pulmonary disease, dementia, or untreatable cancer.
- Advance care planning with the completion of advance directives is especially important for dialysis patients. About half of the dialysis population is elderly. The elderly population have the shortest life expectancy on dialysis and are the most likely to withdraw from dialysis. Prior discussion of advance directives has been shown to help dialysis patients and their families approach death in a reconciled fashion.
- The concept of a “time-limited trial” of dialysis is a way to avoid a sense of inflexible commitment to a treatment strategy when the benefit of the treatment to the particular patient is uncertain. Dialysis may be initiated with a planned reevaluation of whether goals are being met with continuation or withdrawal based on this assessment.
- Prior to implementation of the 2014 United Network for Organ Sharing Kidney Allocation System, troubling disparities had been noted in the kidney transplant program. Studies had found that blacks were less likely than whites to be deemed appropriate candidates for transplantation, to be referred for transplant evaluation, to complete a transplant evaluation, to be added to a transplant waiting list, and to receive a kidney transplant.
- Cherry picking, patient solicitation, and other forms of acting in favor of physician or medical group interest rather than patient welfare compromise a physician’s fiduciary responsibility and the very foundation of medicine and the doctor–patient relationship.
- Because ESRD Seamless Care Organizations are financed based on a shared risk model, large dialysis organizations have a strong incentive to cherry pick healthy patients because a large percentage of their income is at stake. Older and sicker dialysis patients require more dialysis center staff time and resources and are more frequently hospitalized. Both detract from a large dialysis organization’s income in a shared risk model. There is a need for nephrologists and large dialysis organizations to exercise professional ethics and maintain their fiduciary obligation to all patients including those who are the sickest.

PAST: THE EVOLUTION OF DIALYSIS AND TRANSPLANT ETHICS IN THE UNITED STATES

NEPHROLOGY'S CONTRIBUTION TO THE BIRTH OF BIOETHICS

Any review of the ethical issues facing nephrology should start by acknowledging the fundamental role that nephrology played in the early development of the field of bioethics. Because the life-sustaining therapies of dialysis and kidney transplantation predated many parallel therapies in other fields, the ethical issues that inevitably accompany these highly specialized medical treatments were first addressed by nephrologists. This chapter focuses on ethical developments with regard to dialysis and transplantation in the United States, because many innovations and associated ethical challenges occurred first in the United States. These experiences were instructive to other countries. A global perspective on the ethical challenges in treating patients with advanced kidney disease in low-resource countries and those with differing health-care systems is addressed later in this chapter.

Although dialysis had been used to save lives from acute kidney injury for years, chronic dialysis was only possible in 1960 with the invention of the arteriovenous shunt. Amid the excitement of this substantial medical advance, nephrologists were quickly confronted by the unprecedented ethical problem of determining which patients should be granted access to chronic hemodialysis (HD) given the limited availability of expensive machines. It was apparent that this decision required more than medical facts, and that value judgments would have a large influence. The Seattle Artificial Kidney

Center invited a group of citizens to make allocation decisions, a role that was previously well within the guarded realm of physician autonomy. Historian David Rothman heralded this engagement of the lay population as the event that signaled the entrance of bioethics into medicine.¹ Bioethics scholar Albert Jonsen similarly believed that the field of bioethics emerged in response to the nationwide controversy that ensued.²

In his 1964 presidential address to the American Society of Artificial Internal Organs, Dr. Belding Scribner, one of the early fathers of nephrology, identified four major ethical problems that he and his colleagues were facing: 1. patient selection for dialysis; 2. termination of dialysis, which he called “dialysis suicide,” but is now referred to as “stopping” or “withdrawing” dialysis; 3. “death with dignity,” which involved the treatment of patients with end-stage renal disease (ESRD) who were having dialysis withheld or withdrawn; and 4. donor selection for transplantation.³ Dr. Scribner anticipated many of the ethical challenges that would face nephrology and in parallel, many other fields of medicine. Other challenges were yet to be conceived.

Ethical issues arising in treating patients with advanced chronic kidney disease (CKD) can be unpacked using a principlist structure. Principlism is a widely accepted framework for identifying and analyzing ethical problems by applying four ethical principles: respect for autonomy, beneficence, nonmaleficence, and justice.⁴ Fig. 82.1 presents a time line for the relative salience of these principles throughout the history of the ESRD program.⁴ The fourth principle, justice, is particularly relevant to the first ethical challenge that Dr. Scribner identified in his address, patient selection for dialysis, and questions of distributive justice continue to arise in other settings through the evolution of nephrology health care.

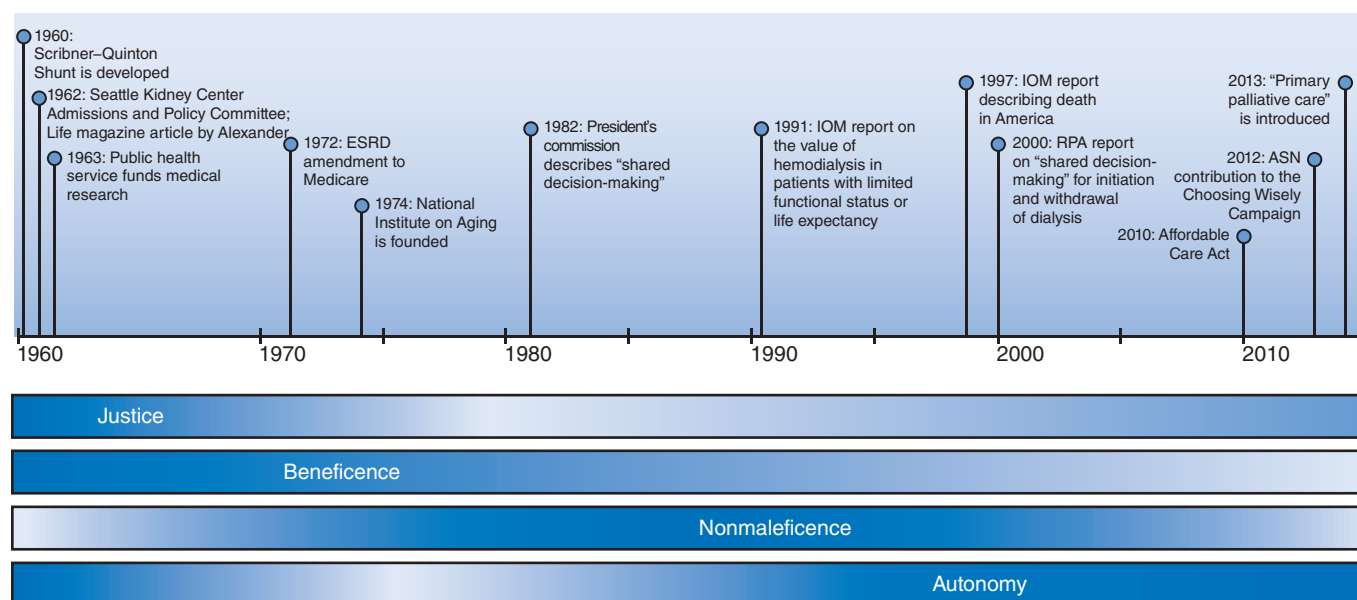


Fig. 82.1 Time line of important events and publications related to the development of dialysis. Although there is significant overlap, the relative salience of each of the four principles varies over this course. ASN, American Society of Nephrology; ESRD, end-stage renal disease; IOM, Institute of Medicine; RPA, Renal Physicians Association. Used with permission of the American Society of Nephrology. Prior publication in the *Clinical Journal of the American Society of Nephrology*: Butler CR, Mehotra R, Tonelli MR, Lam DY. The evolving ethics of dialysis in the United States: A principlist bioethics approach. *Clin J Am Soc Nephrol*. 2016;11:704–709.

RATIONING OF HEMODIALYSIS AND PATIENT SELECTION CRITERIA AT THE BEGINNING OF CHRONIC DIALYSIS IN THE UNITED STATES

The invention of the Quinton–Scribner arteriovenous shunt expanded the patient population who could potentially benefit from HD. When the first US chronic dialysis center, the Seattle Artificial Kidney Center, opened in 1962, the number of patients seeking dialysis treatment far outweighed the center's ability to provide it. Physicians used medical criteria to narrow the group to otherwise healthy patients aged 18 to 45 years, but there remained more candidates than could be accommodated and no established criteria to decide who amongst them would receive the life-sustaining treatment. The ethical principle of justice requires distribution of benefits and burdens in a way that is most fair and/or useful, and that each patient receives that to which she is entitled. In 1962 the ethical problem confronting the Seattle Artificial Kidney Center was one of distributive justice, because they did not have enough dialysis machines or trained personnel to provide dialysis to all patients with ESRD.

Health-care rationing is when a treatment that may benefit a patient is withheld because of resource limitation. The Seattle Artificial Kidney Center was challenged with how to ration dialysis. In response, the Seattle group appointed the Admissions and Policy Committee. The group was composed of a dozen citizens intended to represent a broad swath of the local community and who were tasked with selecting between medically suitable dialysis candidates in the state of Washington. The committee considered choosing patients by lottery, but rejected this idea, believing that more ethical decisions could be made by active selection based on a patient's ability to be rehabilitated and eventually return on society's investment in their health. Factors taken into consideration included age and sex, marital status and number of dependents, income, net worth, emotional stability, educational background, occupation, and future potential contribution. As the selection process evolved, a pattern emerged in which committee decisions weighed heavily on a narrow definition of a favorable personal character and valuable contribution to society.⁵

Journalist Shana Alexander called national attention to the committee in her *Life Magazine* article, "They Decide Who Lives, Who Dies: Medical Miracle Puts Moral Burden on Small Committee."⁵ Once public, the Seattle dialysis patient selection process was subjected to harsh criticism. The committee was castigated for using upper middle-class values and social-worth criteria to make decisions.⁶ Critics wrote, "The prejudices and mindless clichés that pollute the committee's deliberations ... [rule out] creative non-conformists, who rub the bourgeoisie the wrong way but who historically have contributed so much to the making of America. The Pacific Northwest [where Seattle is located in the US] is no place for a Henry David Thoreau with bad kidneys."⁷ The increasingly complex ethical structure on which medicine is practiced was exposed to the public, making several points clear. First, the public has a moral stake in how medical decisions are made. Further, in situations requiring rationing, it is unacceptable to make life and death decisions based on a narrow definition of social worth, though a clear set of guidelines was yet to be developed. Policy decisions over the following years emphasized the country's preference to avoid rationing altogether if possible, even at great financial cost.

US HEALTH-CARE COVERAGE FOR DIALYSIS AND AN EXPANDING DIALYSIS PATIENT DEMOGRAPHIC

The prevailing distaste for health-care rationing practices was reflected by Senator Russell Long of Louisiana following contentious policy debates in the United States, who remarked, "We are the greatest nation in the world, the wealthiest per capita. Are we so hard pressed that we cannot pay for...a life extended for 10–15 years?"⁸ In 1972, passage of the ESRD Amendment to Medicare, Section 299I of Public Law 92-603, virtually eliminated the need to ration dialysis. The US Congress passed this legislation because of the clear life and death urgency of chronic dialysis for patients with ESRD and the expectation that the number of patients who would need this treatment would be limited. This legislation classified ESRD patients who were otherwise eligible for Social Security as disabled and authorized a Medicare entitlement covering dialysis and transplantation.⁹ Neither Congress nor physicians thought it necessary or proper for the government to further direct which patients were appropriate candidates for dialysis. Although in the early days of dialysis, medical criteria for acceptance to the dialysis program had limited patients to those who were young and relatively healthy, many believed that it was morally unjustified to deny dialysis treatment to any patient with ESRD.⁶ A decade later, the first report of the US Renal Data System documented the progressively greater acceptance rate of patients for dialysis treatment,¹⁰ explained in part by the inclusion of patients who had poor prognoses, especially the elderly and those with diabetic nephropathy.¹¹ By 2000, 48% of new patients starting dialysis were 65 years of age or older and 45% had ESRD from diabetes. Observers raised concerns about the appropriateness of treating patients with a limited life expectancy and quality of life.^{9,12} Others questioned the utility of providing dialysis to patients with life-threatening illnesses such as end-stage cardiovascular disease, debilitating pulmonary disease, or untreatable cancer. Benefit was also unclear in patients with severe neurologic disease which rendered them unable to engage with others, such as those in a persistent vegetative state, with severe dementia, or with debilitating cerebrovascular disease.¹³

In 1991, the Institute of Medicine Committee for the Study of the Medicare ESRD Program emphasized that the existence of the public entitlement for treatment of ESRD does not obligate physicians to treat all patients who have kidney failure with dialysis or transplantation.⁹ For some, the burden of dialysis substantially outweighed the benefit. The provision of dialysis to these patients would violate the medical maxim, "Be of benefit and do no harm." In other words, the decision to institute a life-sustaining therapy required a balance of beneficence and nonmaleficence. Systematic differences amongst providers and regional variation in gauging the resultant balance of these factors risked causing health-care disparity.^{14,15} The committee recommended that guidelines be developed for identifying such patients but with continued respect for physicians' discretion in individual assessment. Such guidelines were intended to help nephrologists make dialysis decisions more uniformly, with greater ease, and in a way that promoted patient benefit. The first edition of the clinical practice guidelines on *Shared Decision-Making in the Appropriate Initiation of and Withdrawal from Dialysis* was published in 2000.¹⁶

WITHDRAWAL FROM DIALYSIS

In the 1960s, patients felt fortunate to be among the few chosen by dialysis selection committees to receive a rationed therapy. No one anticipated that a patient might choose to forgo this sought-after treatment knowing that they would shortly die. Practitioners grappled with whether this so-called dialysis suicide was a form of psychopathology or whether, given the unprecedented nature of the life-sustaining therapy, requests to “stop the machine” should be interpreted differently. Physicians and ethicists drew on Catholic moral theology that makes a distinction between ordinary and extraordinary measures, the latter not being morally obligatory. They concluded that patients may refuse or stop aggressive interventions such as dialysis, because they represent extraordinary attempts to sustain life.¹⁷

In its 1983 report, *Deciding to Forego Life-Sustaining Treatment*, the President’s Commission for the Study of Ethical Problems in Medicine and Biomedical and Behavioral Research considered ethical and legal issues raised by life-sustaining treatments such as dialysis. The commission understood that “biomedical developments ... have made death more a matter of deliberate decision. ... Matters once the province of fate have now become a matter of human choice.”¹⁸ The President’s Commission reached two major conclusions that provide justification for withholding and withdrawing dialysis: “the voluntary choice of a competent and informed patient should determine whether or not life-sustaining therapy will be undertaken; and health-care professionals serve patients best by maintaining a presumption in favor of sustaining life, while recognizing that competent patients are entitled to choose to forgo any treatments, including those that sustain life.”¹⁸

Similarly, in its 1993 annual report, the End-Stage Renal Disease Data Advisory Committee to the US Renal Data System articulated ethical justifications and general and specific principles to be used in making decisions about withholding or withdrawing dialysis. This report reflected the deliberations of an ad hoc committee composed of nephrologists, ethicists, and health policy experts. The report described two ethical justifications for forgoing dialysis: 1. the right of patients to refuse dialysis based on the ethical principle of respect for autonomy and the legal right of self-determination; and 2. a judgment that dialysis does not offer a reasonable expectation of medical benefit based on the ethical principles of beneficence and nonmaleficence.¹⁹

An early study attempting to describe how these principles were adopted in practice noted that dialysis withdrawal accounted for 22% of deaths in one large dialysis program,²⁰ ranking behind cardiovascular disease and infection as the most common cause of death in this population. Subsequent research found that older patients (those aged 65 years or older) and patients with diabetes were the most likely to stop dialysis. According to a survey of dialysis units performed in 1990, at least one and on average three patients withdrew from dialysis in the preceding year.¹⁴

The recommendations of these regulatory bodies emphasize the importance of optimizing benefit and reducing harm while introducing the emerging concept of respect for autonomous choice. However, the process of arriving at a balance between these values incorporating

input from the nephrologist, patient, and family remained opaque. Simple answers were not forthcoming as individual patients evaluate benefits and burdens differently and enter discussion with varying cognitive abilities and social backgrounds.

PRESENT

ADVANCE DIRECTIVES AND DIALYSIS PATIENTS

By the 1990s, studies showed that 90% of nephrologists would stop dialysis at the request of a patient with decision-making capacity.^{14,15} Nephrologists also expressed a willingness to stop dialysis of “irreversibly incompetent” patients who had previously documented their wish to stop in such a condition. However, they disagreed about how to treat these patients in the absence of clear advance directives.¹⁵ Following a precedent set in the earliest days of dialysis, there was a strong default toward continued treatment for patients who could not or had not expressed their wishes. Therefore, a patient’s right to forgo dialysis was difficult to exercise in these circumstances.

The Patient Self-Determination Act, which required institutions participating in Medicare and Medicaid to educate patients about the importance of completing advance directives, became effective in December 1991.²¹ Although the ESRD program was almost entirely funded by Medicare, dialysis units were inadvertently left out of the law. This was a problematic omission given that advance care planning is especially important for dialysis patients for several reasons²²: 1. about half of the dialysis population is elderly, and the elderly have the shortest life expectancy on dialysis and are the most likely to withdraw or be withdrawn from dialysis; 2. prior discussion of advance directives has been shown to help dialysis patients and their families approach death in a reconciled fashion²³; 3. patients who discuss and complete written advance directives are significantly more likely to have their wish to die at home respected; 4. unless a specific directive to withhold cardiopulmonary resuscitation is obtained—which can be done in the framework of advance care planning—it will be automatically provided, even though it rarely leads to extended survival in dialysis patients.²⁴

It had already been established that many dialysis patients have a personal threshold at which point they would choose to stop aggressive care. Discussions and documentation to specify these situations in advance protect autonomy and personal preferences when patients are unable to advocate for themselves. For these reasons, nephrologists have been encouraged to discuss the circumstances under which patients would want to discontinue dialysis and forgo cardiopulmonary resuscitation and to urge patients to complete written advance directives.²⁵ When patients lack decision-making capacity and have not completed advance directives, ethically complex issues often arise in discussion about whether to start or stop dialysis. In these situations, many nephrologists indicate that they would consult an ethics committee for assistance.¹⁴ See Chapter 62 for more discussion on the practical aspects of decision-making in this context.

FINANCING DIALYSIS CENTERS AND CONFLICTS OF INTEREST

Conflict of interest occurs when there is a clash between a physician or medical group's interests and the welfare of a patient. These conflicts represent a serious challenge to a physician–patient relationship, which is based on open disclosure of patient information under the assumption of a physician's fiduciary responsibility to prioritize the patient's goals over their own. Fee-for-service medicine will always involve some conflict of interest, but nephrology has encountered particularly problematic situations when physicians share ownership in for-profit dialysis centers in which they treat patients. In these cases, nephrologists are financially rewarded for reducing costs. While not inherently problematic, cost reduction can compromise patient welfare if it entails decreasing dialysis time, purchasing less effective equipment, or hiring fewer nurses, social workers, and dietitians.

For-profit dialysis companies flourished under the early ESRD program and, by 1990, they accommodated two-thirds of ESRD patients.²⁶ Physicians were paid per-patient and centers paid per-treatment. Although this incentive structure does not necessitate unethical practice, trends in care suggested that compensation may have driven some aspects of care. The United States quickly developed the highest dialysis patient acceptance rate in the world.¹¹ Short dialysis times, which are associated with increased mortality, were found disproportionately in for-profit dialysis units. Patients treated in these units had a 20% higher mortality rate and a 26% lower rate of referral for renal transplantation.²⁷

In order to hold dialysis units accountable for the quality of care they provided, the Medicare ESRD program, through the 18 ESRD networks, established quality indicators to measure the performance of dialysis units both individually and regionally. Metrics included adequacy of dialysis, anemia management, vascular access placement, and standardized mortality ratios. Despite the intention to improve care, some raised concerns that a focus on metrics could lead practitioners to “cherry pick” healthier patients who would improve their units' aggregate numbers while restricting access to sicker or more socially complex patients.²⁸ Cherry picking, patient solicitation, and other forms of acting in favor of physician or medical group interest rather than patient welfare compromise a physician's fiduciary responsibility and the very foundation of medicine and the doctor–patient relationship.

CKD CARE AND DIALYSIS OF THE OLDER AND/OR MULTIMORBID PATIENT

Old age and medical comorbidity are no longer seen as contraindications to dialysis. Far from it, as the population ages and diabetes and other chronic conditions become more prevalent, the number of very elderly patients (those aged 75 years or older) has become the fastest growing patient population on dialysis.²⁹ Deciding whether to start elderly patients with CKD on dialysis involves an understanding and consideration of factors common to this group including reduced life span, complex comor-

bidity, heterogeneous personal values, and varied quality of life.³⁰

Between 1996 and 2003, rates of dialysis initiation among octogenarians and nonagenarians nearly doubled.³¹ At the same time, median survival for patients aged 85 to 89 beginning dialysis was only 8.4 months compared with 24.9 months for patients aged 65 to 70.³¹ Clinical characteristics such as older age, nonambulatory status, poor nutritional status, and presence of comorbid conditions were associated with shorter life span.³¹ Amongst US veterans with CKD stage 3 to 5, those older than 75 were far more likely to die than to develop ESRD.³² In a study comparing dialysis with conservative treatment in a population of patients greater than 75 years old with advanced CKD, there was a 2.9-fold greater survival on dialysis.³⁰ However, in a multivariate subgroup analysis of these older patients, dialysis offered no survival benefit for patients with ischemic heart disease or those with the highest comorbidity scores. This finding has since been replicated.^{33,34} Moreover, researchers have found that comorbid conditions do not influence the decision to initiate dialysis treatment despite the fact that comorbidity strongly influences CKD patients' prognosis.³⁵

Trends toward higher medical resource use among the older population despite uncertain benefits have led some to be concerned about the “biomedicalization of aging” in which technology is used for its own sake rather than as merely a means to support individual life goals.³⁶ Medical anthropologist Sharon Kaufman identified indiscriminant use of dialysis in the elderly as a prime example. She describes the “routinization” of starting dialysis for older patients, and the impact of societal expectations in which there is no age limit to the use of standard medical treatments. She writes that nephrologists report they feel obligated to dialyze very old patients even if they have personal moral concerns about whether the patient is likely to benefit.³⁷

Given the unique aspects of caring for an older and multimorbid population, the Renal Physicians Association and American Society of Nephrology clinical practice guideline emphasized the importance of discussing prognosis and advance care planning with elderly patients and their families to make individualized decisions.^{16,31} These data argue against an “age-neutral” approach to the management of CKD and emphasized the need for prognostic tools that enable clinicians to identify the subgroup of older patients who are most likely to benefit from dialysis.³² An integrated prognostic tool to estimate 6-month mortality in dialysis patients was developed and validated with a C-statistic of 0.80. It considers age, serum albumin, dementia, peripheral vascular disease, and the nephrologist's response to the surprise question (“Would I be surprised if this patient died in the next six months?”)³⁸ The accuracy of this tool in the stage 5 CKD population remains to be studied. One observer raised the possibility that the standard content of informed consent for dialysis warrants an age-sensitive approach that is attuned to the different balance of advantages and harms of dialysis for older patients with multiple comorbidities.³⁹ The American Geriatrics Society has developed a structure for approaching the medical evidence base recognizing that it is often difficult to extrapolate results from trial data to an elderly population.⁴⁰ See Chapter 84 for more discussion of decision-making and clinical management of elderly patients with CKD.

Clinical Relevance

Clinical characteristics of advanced chronic kidney disease patients such as older age, nonambulatory status, poor nutritional status, and presence of comorbid conditions are associated with shorter life span. In a multivariate subgroup analysis of older patients, dialysis offered no survival benefit for patients with ischemic heart disease or those with the highest comorbidity scores.

ETHICAL ISSUES IN ESRD END-OF-LIFE CARE

The Robert Wood Johnson Foundation ESRD Peer Work Group stated in 2002 that “most patients with ESRD, especially those who are not candidates for renal transplantation, have a significantly shortened life expectancy.”⁴¹ In the wake of public dissatisfaction with end-of-life care practices across the breadth of chronic disease, nephrology physician groups, including the Renal Physicians Association and American Society of Nephrology, recognized that they have a particular ethical responsibility to improve patients’ experience of the end of life. Given the shortened life expectancy and high symptom burden of this population, they noted it was especially important that nephrologists and dialysis staff learn primary palliative care techniques and also have the ability to refer to palliative specialists. Nephrology fellowships generally have yet to respond to this need with specialized training in primary palliative care and end-of-life care, but trainees report a sense of moral obligation to provide such care.⁴²

Palliative care strategies can promote the autonomy of CKD patients. Although notoriously difficult for clinicians, prognostication is critically important for patients planning their remaining time. The “surprise” question identifies dialysis patients at high risk for early mortality. The odds of a patient dying within a year were shown to be 3.5 times higher if the nephrologist or nephrology nurse practitioner answered “No” than for those patients about whom the provider answered “Yes.”⁴³ Nephrologists have been encouraged to implement the surprise question monthly on dialysis rounds to screen for patients who might benefit from palliative care intervention.

When dialysis is deemed not beneficial, it is critical to have alternative care options available. Physicians struggle with the decision against dialysis when there seems to be “nothing else to offer.” Patients similarly feel that they are not adequately counseled on alternatives to dialysis and express a sense of “powerlessness” when faced with ESRD.⁴⁴ “Comprehensive conservative care” as defined by a Kidney Disease Improving Global Outcomes (KDIGO) consensus group includes interventions to delay progression, control symptoms, and promote social support, but excludes treatment with dialysis.⁴⁵ In order for this approach to be viable, further research and evidence are needed, as well as established pathways to mirror the multidisciplinary coordinated care that is already built into the dialysis network (see Chapter 62: Supportive Care in Advanced Chronic Kidney Disease).

Even with effective communication and preparation, it is often difficult for patients to fully anticipate the impact of dialysis on their life. The concept of a “time-limited trial” is a way to avoid a sense of inflexible commitment to a treatment strategy when the benefit of the treatment to the particular

patient is uncertain.⁴⁶ Dialysis may be initiated with a planned reevaluation of whether goals are being met with continuation or withdrawal based on this assessment.

SHARED DECISION-MAKING AND CLINICAL PRACTICE GUIDELINE ON APPROPRIATE USE OF DIALYSIS

In an effort to reduce the variation in dialysis practices between physicians and across regions of the United States, clinical practice guidelines have been established over the last decades. By the early 1990s, demographics of the ESRD population had changed greatly from earlier years. Nephrologists reported being asked to provide dialysis to patients for whom it seemed of marginal benefit including patients who were elderly and with multiple comorbidities. The US Renal Data System confirmed highest incident treatment rates for 75- to 79-year-old patients.⁴⁷ By contrast, research found that some patients regret their decision to start dialysis, but felt they had no other choice.^{48,49} In 1991, the Institute of Medicine recommended development of an evidence-based, clinical practice guideline “for evaluating patients for whom the burdens of renal replacement therapy may substantially outweigh the benefits.”¹³

A paradigm of “shared decision-making” was developed to address these questions, and has become the ethical standard for health-care practice. Through this process, physicians and patients agree on a specific course of action based on a common understanding of treatment goals and the risks and benefits of each option. However, there are acknowledged limits to this process in cases where the patient’s preferences and medical team’s sense of beneficial course cannot be reconciled. In these cases, a patient has the right to refuse dialysis even if the renal care team disagrees with the decision. Similarly, the renal care team has the right to refuse to offer dialysis when the expected benefits do not justify the risks.

In 2000 the Renal Physicians Association and the American Society of Nephrology published the clinical practice guideline *Shared Decision-Making in the Appropriate Initiation of and Withdrawal from Dialysis*.¹⁶ These guidelines were intended to help those involved in care of CKD and ESRD patients including nephrologists, intensivists, primary-care physicians, nephrology nurses, advanced practice nurses, and nephrology social workers. The guidelines are also potentially useful to patients and their families, renal dietitians, dialysis technicians, renal administrators, clergy, and policy makers.

In the following years, an extensive evidence base was developed regarding dialysis decision-making, including 1. the poor prognosis of some elderly stage 4 and 5 CKD patients, many of whom are likely to die prior to initiation of dialysis or for whom dialysis may not provide a survival advantage over medical management without dialysis; 2. the prevalence of cognitive impairment in dialysis patients and the need to periodically assess them for decision-making capacity; 3. the underrecognition and undertreatment of pain and other symptoms in dialysis patients; 4. the underutilization of hospice in dialysis patients; and 5. the distinctly different treatment goals of ESRD patients based on their overall condition and personal preferences. In 2010 the Renal Physicians Association developed a second edition of the guideline to provide

Box 82.1 Systematic Evaluation of a Patient or Family Request to Stop Dialysis⁵⁰

1. Determine the reasons or conditions underlying the patient/surrogate desires regarding withdrawal of dialysis. Such assessment should include specific medical, physical, spiritual, and psychological issues, as well as interventions that could be appropriate. Some of the potentially treatable factors that might be included in the assessment are as follows:
 - a. Underlying medical disorders, including the prognosis for short- or long-term survival on dialysis
 - b. Difficulties with dialysis treatments
 - c. The patient's assessment of his or her quality of life and ability to function
 - d. The patient's short- and long-term goals
 - e. The burden that costs of continued treatment/medications/diet/transportation may have on the patient/family/others
 - f. The patient's psychological condition, including depression and conditions or symptoms that may be caused by uremia
 - g. Undue influence or pressure from outside sources, including the patient's family
 - h. Conflict between the patient and others
 - i. Dissatisfaction with the dialysis modality, the time, or the setting of treatment
2. If the patient wishes to withdraw from dialysis, did he or she consent to referral to a counseling professional (e.g., social worker, spiritual advisor, psychologist, or psychiatrist)?
3. If the patient wishes to withdraw from dialysis, are there interventions that could alter the patient's circumstances that might result in his or her considering it reasonable to continue dialysis?
 - a. Describe possible interventions.
 - b. Does the patient desire the proposed intervention(s)?
4. In cases in which the surrogate has made the decision to either continue or withdraw dialysis, has it been determined that the judgment of the surrogate is consistent with the stated desires of the patient?
5. Questions to consider when a patient asks to stop dialysis:
 - a. Is the patient's decision-making capacity diminished by depression, encephalopathy, or other disorder?
 - b. Why does the patient want to stop dialysis?
 - c. Are the patient's perceptions about the technical or quality-of-life aspects of dialysis accurate?
 - d. Does the patient really mean what he or she says or is the decision to stop dialysis made to get attention, help, or control?
 - e. Can any changes be made that might improve life on dialysis for the patient?
 - f. Would the patient be willing to continue dialysis while the factors responsible for the patient's request are addressed?
 - g. Has the patient discussed his or her desire to stop dialysis with significant others such as family, friends, or spiritual advisors? What do they think about the patient's request?

From Shared Decision-Making in the Appropriate Initiation of and Withdrawal From Dialysis. 2nd ed. Rockville, MD: Renal Physicians Association; 2010.

clinicians, patients, and families with 1. the most current evidence about the benefits and burdens of dialysis for patients with diverse conditions; 2. recommendations for quality in decision-making about treatment of patients with acute kidney injury, CKD, and ESRD; and 3. practical strategies to help clinicians implement the guideline recommendations.⁵⁰ **Box 82.1** presents a practical approach from the guideline on how to systematically evaluate a patient's request to stop dialysis.

This second edition also made explicit recommendations to integrate palliative care into the treatment of CKD and ESRD patients and to develop quality metrics for dialysis patients whose goals of care focus on comfort that are distinct from those whose goals are to aggressively prolong life and function. It noted that good communication improves patients' adjustment to illness, increases adherence to treatment, and results in higher patient and family satisfaction. The second edition called on nephrologists and other members of the interdisciplinary team to communicate prognostic information, allowing patients to reach more informed decisions about dialysis and transplantation options. It urged nephrology fellowship programs to incorporate training in effective, empathetic communication skills so that patients' decisions can be based on an accurate understanding of their overall condition and the pros and cons of treatment options.⁵⁰ The roles of supportive or palliative care and incorporation of these options into treatment planning are discussed further in Chapter 62.

The objectives for both editions of this guideline are listed in **Box 82.2**. The second edition provides 10 recommendations with regard to decision-making about withholding or withdrawing dialysis and the care of patients who forgo dialysis

Box 82.2 Objectives of the Clinical Practice Guideline Shared Decision-Making in the Appropriate Initiation of and Withdrawal From Dialysis⁵⁰

- Synthesize available research evidence regarding patients with acute kidney injury and end-stage renal disease (ESRD) as a basis for making recommendations about withholding and withdrawing dialysis.
- Enhance understanding of the principles and processes useful for and involved in making decisions to withhold or withdraw dialysis.
- Promote ethically as well as medically sound decision-making in individual cases.
- Recommend tools that can be used to promote shared decision-making in the care of patients with acute kidney injury or ESRD.
- Offer a publicly understandable and acceptable ethical framework for shared decision-making among health-care providers, patients, and their families.

From Shared Decision-Making in the Appropriate Initiation of and Withdrawal From Dialysis. 2nd ed. Rockville, MD: Renal Physicians Association; 2010.

(**Box 82.3**). These guidelines have been widely quoted in the nephrology and palliative care literature, and studies have documented their effectiveness.^{51–53} More than half of US nephrologists in 2005 reported awareness and use of these guidelines and surveys, and a comparison of nephrologist

Box 82.3 Renal Physicians Association Clinical Practice Guideline Recommendations for Shared Dialysis Decision-Making⁵⁰

Establishing a Shared Decision-Making Relationship

Recommendation No. 1

Develop a physician–patient relationship for shared decision-making.

Shared decision-making is the recognized preferred model for medical decision-making because it addresses the ethical need to fully inform patients about the risks and benefits of treatments, as well as the need to ensure that patients' values and preferences play a prominent role. Because of the number and complexity of decisions involved in treating kidney failure, a shared decision-making relationship is particularly important for patients with acute kidney injury (AKI), stage 4 and 5 chronic kidney disease (CKD), and stage 5 CKD requiring dialysis end-stage renal disease (ESRD). Participants in shared decision-making should involve at a minimum the patient and the physician. In addition, patients should identify and include a person who could serve as their decision-maker in the event they lose decision-making capacity. If a patient lacks decision-making capacity, decisions should involve the person legally authorized to make health-care decisions on behalf of the incapacitated patient. This person is often (though not always) a family member and will be called "the legal agent" in the remainder of this document (see Glossary for a full description). With the patient's consent, shared decision-making may include family members or friends and other members of the health-care team.

Informing Patients

Recommendation No. 2

Fully inform AKI, stage 4 and 5 CKD, and ESRD patients about their diagnosis, prognosis, and all treatment options.

In the setting of critical illness many patients with CKD will require urgent dialysis and the vast majority of patients with AKI will have multiple medical problems, in addition to kidney failure. The concept of shared decision-making necessitates a multidisciplinary approach including nephrologists, intensivists, and others as appropriate and decisions about acute renal replacement therapy should be made in the context of other life-sustaining treatments. For example, a decision to withhold dialysis in a patient agreeing to and receiving multiple other forms of life-sustaining therapy could represent discordant treatment in the same way that offering dialysis to a patient who has decided to forgo other forms of life-sustaining therapy might be inappropriate. Intensive care physicians need to be included in shared decision-making for kidney patients in the intensive care unit (ICU).

For ESRD patients, the shared decision-making options include:

1. available dialysis modalities and kidney transplantation if applicable
2. not starting dialysis and continuing medical management
3. a time-limited trial of dialysis
4. stopping dialysis and receiving end-of-life care.

Choices among options should be made by patients or, if patients lack decision-making capacity, their designated legal agents. Their decisions should be informed and voluntary. The renal care team, in conjunction with the primary care physician, should insure that the patient or legal agent understands the benefits and burdens of dialysis and the consequences of not starting or stopping dialysis. Research studies have identified a population of CKD patients for whom the prognosis is particularly poor. This population has been found to include patients with two or more of the following characteristics: (1) elderly (defined by research studies identifying

poor outcomes in patients who are aged 75 years and older); (2) patients with high comorbidity scores (e.g., modified Charlson Comorbidity Index score of ≥ 8); (3) marked functional impairment (e.g., Karnofsky Performance Status Scale score of < 40); and (4) severe chronic malnutrition (e.g., serum albumin level < 2.5 g/dL using the bromocresol green method). Patients in this population should be informed that dialysis may not confer a survival advantage or improve functional status over medical management without dialysis and that dialysis entails significant burdens that may detract from their quality of life.

Recommendation No. 3

Give all patients with AKI, stage 5 CKD, or ESRD an estimate of prognosis specific to their overall condition.

To facilitate informed decisions about starting dialysis for AKI, stage 5 CKD, or ESRD, all patients should have their prognosis estimated and discussed, with the realization that the ability to predict survival in the individual patient is limited. Depending on the setting, a primary care physician, intensivist, or nephrologist who is familiar with estimating and communicating prognosis should conduct these discussions (see [Recommendation No. 10](#) for communication strategies). For patients with ESRD, the "surprise" question "Would I be surprised if this patient died in the next year?" can be used together with known risk factors for poor prognosis: age, comorbidities, severe malnutrition, and poor functional status. For patients with stage 5 CKD predialysis, the estimate of prognosis should be discussed with the patient or legal agent, patient's family, and among the medical team members to develop a consensus on the goals of care and whether dialysis or active medical management without dialysis should be used to best achieve these goals. These discussions should occur as early as possible in the course of the patient's kidney disease and continue as the kidney disease progresses. For ESRD patients on dialysis who experience major complications that may substantially reduce survival or quality of life, it is appropriate to reassess treatment goals, including consideration of withdrawal from dialysis.

Facilitating Advance Care Planning

Recommendation No. 4

Institute advance care planning.

The purpose of advance care planning is to help the patient understand his/her condition, identify his/her goals for care, and prepare for the decisions that may have to be made as the condition progresses over time. For chronic dialysis patients, the interdisciplinary renal care team should encourage patient–family discussion and advance care planning and include advance care planning in the overall plan of care for each individual patient (see Section 10: Glossary for definition of renal care team). The renal care team should designate a person to be primarily responsible for ensuring that advance care planning is offered to each patient. Patients with decision-making capacity should be "strongly" encouraged while they have capacity to talk to their legal agents to ensure that the legal agent knows the patient's wishes and agrees to make decisions according to these wishes.

The renal care team should attempt to obtain written advance directives from all dialysis patients. Where legally accepted, Physician Orders for Life-Sustaining Treatment or similar state-specific forms, also should be completed as part of the advance care planning process. At a minimum, each dialysis patient should be asked to designate a legal agent in a state-specific advance directive. Advance directives should be honored by dialysis centers,

Continued on following page

Box 82.3 Renal Physicians Association Clinical Practice Guideline Recommendations for Shared Dialysis Decision-Making (Continued)

nephrologists, and other nephrology clinicians except possibly in situations in which the advance directive requests treatment contrary to the standard of care (see [Recommendation No. 8](#) on conflict resolution).

Making a Decision to Not Initiate or to Discontinue Dialysis

Recommendation No. 5^a

If appropriate, forgo (withhold initiating or withdraw ongoing) dialysis for patients with AKI, CKD, or ESRD in certain, well-defined situations.

These situations include the following:

- Patients with decision-making capacity, who being fully informed and making voluntary choices, refuse dialysis or request that dialysis be discontinued.
- Patients who no longer possess decision-making capacity who have previously indicated refusal of dialysis in an oral or written advance directive.
- Patients who no longer possess decision-making capacity and whose properly appointed legal agents/surrogates refuse dialysis or request that it be discontinued.
- Patients with irreversible, profound neurological impairment such that they lack signs of thought, sensation, purposeful behavior, and awareness of self and environment.

Recommendation No. 6

Consider forgoing dialysis for AKI, CKD, or ESRD patients who have a very poor prognosis or for whom dialysis cannot be provided safely.

Included in these categories of patients are the following:

- Those whose medical condition precludes the technical process of dialysis because the patient is unable to cooperate (e.g., advanced dementia patient who pulls out dialysis needles) or because the patient's condition is too unstable (e.g., profound hypotension).
- Those who have a terminal illness from nonrenal causes (acknowledging that some in this condition may perceive benefit from and choose to undergo dialysis).
- Those with stage 5 CKD older than age 75 years who meet two or more of the following statistically significant very poor prognosis criteria (see [Recommendation Nos 2 and 3](#)): 1. clinicians' response of "No, I would not be surprised" to the surprise question; 2. high comorbidity score; 3. significantly impaired functional status (e.g., Karnofsky Performance Status score < 40); and 4. severe chronic malnutrition (i.e., serum albumin < 2.5 g/dL using the bromocresol green method).

Resolving Conflicts About What Dialysis Decisions to Make

Recommendation No. 7

Consider a time-limited trial of dialysis for patients requiring dialysis, but who have an uncertain prognosis, or for whom a consensus cannot be reached about providing dialysis.

If a time-limited trial of dialysis is conducted, the nephrologist, the patient, the patient's legal agent, and the patient's family (with the patient's permission to participate in decision-making) should agree in advance on the length of the trial and parameters to be assessed during and at the completion of the time-limited trial to determine whether dialysis has benefited the patient and whether dialysis should be continued.

Recommendation No. 8

Establish a systematic due process approach for conflict resolution if there is disagreement about what decision should be made with regard to dialysis.

Conflicts may occur between the patient/legal agent and the renal care team about whether dialysis will benefit the patient. Conflicts also may occur within the renal care team or between the renal care team and other health-care providers. In sitting down and talking with the patient/legal agent, the nephrologist should try to understand their views, provide data to support his/her recommendation, and correct misunderstandings. In the process of shared decision-making, the following potential sources of conflict have been recognized: 1. miscommunication or misunderstanding about prognosis; 2. intrapersonal or interpersonal issues; or 3. special values. If dialysis is indicated emergently, it should be provided while pursuing conflict resolution, provided the patient or legal agent requests it.

Providing Effective Palliative Care

Recommendation No. 9

To improve patient-centered outcomes, offer palliative care services and interventions to all AKI, CKD, and ESRD patients who suffer from burdens of their disease.

Palliative care services are appropriate for people who choose to undergo or remain on dialysis and for those who choose not to start or to discontinue dialysis. With the patient's consent, a multiprofessional team with expertise in renal palliative care, including nephrology professionals, family, or community-based professionals, and specialist hospice or palliative care providers, should be involved in managing the physical, psychological, social, and spiritual aspects of treatment for these patients, including end-of-life care. Physical and psychological symptoms should be routinely and regularly assessed and actively managed. The professionals providing treatment should be trained in assessing and managing symptoms and in advanced communication skills. Patients should be offered the option of dying where they prefer, including at home with hospice care, provided there is sufficient and appropriate support to enable this option. Support also should be offered to patients' families, including bereavement support where appropriate. Dialysis patients for whom the goals of care are primarily comfort should have quality measures distinct from patients for whom the goals are aggressive therapy with optimization of functional capacity.

Recommendation No. 10

Use a systematic approach to communicate about diagnosis, prognosis, treatment options, and goals of care.

Good communication improves patients' adjustment to illness, increases adherence to treatment, and results in higher patient and family satisfaction with care. Patients appreciate sensitive delivery of information about their prognosis and the ability to balance reality while maintaining hope. In communicating with patients, the critical task for clinicians is to integrate complicated biomedical facts and conditions with emotional, social, and spiritual realities that are equally complex but not well described in the language of medicine. This information must be communicated in a way that patients, legal agents, and families can understand and use to reach informed decisions about dialysis and transplantation options. Patients' decisions should be based on an accurate understanding of their condition and the pros and cons of treatment options. To facilitate effective communication,

Box 82.3 Renal Physicians Association Clinical Practice Guideline Recommendations for Shared Dialysis Decision-Making (Continued)

reliance upon a multidisciplinary approach including nephrologists, intensivists, and others as appropriate is warranted. Decisions about acute renal replacement therapy in AKI should be made in the context of other life-sustaining treatments. Intensive care physicians should be included in shared decision-making for kidney

patients in the ICU to facilitate discussions on global disease or injury prognosis. Fellowship programs should incorporate training to help nephrologists develop effective, empathetic communication skills, which are essential in caring for this patient population.

^aMedical management incorporating palliative care is an integral part of the decision to forgo dialysis in AKI, CKD, or ESRD, and attention to patient comfort and quality of life while dying should be addressed directly or managed by palliative care consultation and referral to a hospice program (see [Recommendation No. 9](#) on palliative care services).

From *Shared Decision-Making in the Appropriate Initiation of and Withdrawal From Dialysis*. 2nd ed. Rockville, MD: Renal Physicians Association; 2010.

knowledge and attitudes between 1991 and 2005 showed increasing alignment with their recommendations. More homogeneity has been noted in practices on withholding care in patients with permanent cognitive deficits. More US dialysis centers had written policies on cardiopulmonary resuscitation and dialysis withdrawal. Nephrologists were more likely to honor a dialysis patient's do-not-resuscitate order (66% in 1990 and 83% in 2005) and to consider consulting an ESRD network ethics committee (39% in 1990 and 52% in 2005).⁵³ Although these results are promising, recent research continues to show that shared decision-making remains poorly integrated into the practice of predialysis counseling of advanced CKD patients about their treatment options.⁵⁴

Because of the early recognition of heterogeneity in practice causing potential health-care disparity and issues of justice, clinical practice guidelines have focused on practice and process of withdrawing and withholding dialysis. A recent meta-analysis draws attention to the breadth of ethical issues in nephrology with many related to heterogeneous practice including referral from primary care to nephrology, referral to transplantation, delivery of prognosis, and availability of alternative treatment options.⁵⁵ The authors recognize the success of guidelines in dialysis care and suggest that development of a broader scope of clinical practice guidelines may address these growing concerns. Shared-decision making is challenging because it requires robust and recurring discussion of core patient values and treatment options, but if done well, it may significantly improve patient-centered care for dialysis patients.⁵⁴

Clinical Relevance

A paradigm of shared decision-making has become the ethical standard for health-care practice. Through this process, physicians and patients agree on a specific course of action based on a common understanding of treatment goals and the risks and benefits of each option. Shared decision-making prior to the initiation of dialysis has been recommended by the American Society of Nephrology and the Renal Physicians Association.

ACCESS TO RENAL TRANSPLANTATION IN THE UNITED STATES

Dr. Scribner was prescient in 1964 when he identified donor selection for transplantation as one of the major ethical

problems facing nephrologists. There is little disagreement that the quality of life for ESRD patients is often better after transplantation, and most patients prefer this strategy to continuing dialysis.⁵⁶ Research also shows that transplantation is associated with lower costs to the health-care system. In recent times, organ shortage has replaced immunologic barriers as the major hurdle to transplantation.⁵⁷ Because the growth in the number of transplantations has not kept pace with the growth in the incidence of ESRD, the rate of transplantations per 100 dialysis patient-years has actually fallen annually since the 1990s.²⁹ In its 2016 Annual Report, the US Renal Data System indicated that 88,231 people were on the kidney transplant waiting list. In many parts of the country the average waiting time was >5 years.²⁹

Ethical issues arise in both attempts to increase the donor pool and appropriate allocation of available donor kidneys. As was discovered during the early days of dialysis, criteria for appropriate distribution of a limited resource are difficult and can be fraught with the biases of those making allocation decisions. Transplantation carries with it additional ethical issues related to respecting the unique gift of an organ whether from a living or deceased donor.

The majority of transplanted kidneys come from deceased donors. In an attempt to expand this donor pool, new policies allow use of a greater breadth of types of deceased donor kidneys including grafts from older donors and donors after cardiac death. The Institute of Medicine committee recently considered a variety of additional approaches to expand the pool ([Box 82.4](#)).⁵⁸ However, even given a larger number of donors, the medical community is unlikely to close the gap between supply and demand.⁵⁹ This absolute limitation on resources necessitates rationing, and the proper strategy to distribute organs most fairly while also using them to their greatest benefit remains an ongoing debate. At its core, the United Network for Organ Sharing allocation system is a waiting line in which candidates are prioritized by how long they have been listed. This egalitarian strategy intends to give equal weight to all-comers. In an attempt to also satisfy the other value of distributive justice – utility – other criteria match kidneys with patients who may benefit most from them. For example, priority is given to a particularly good immunological match. Although there are benefits and drawbacks to all criteria, they are openly published and available for debate and change. In 2014, the New Kidney Allocation System incorporated a strategy of “longevity matching” in attempts to fit young, healthy recipients with more robust kidneys while matching older recipients with grafts that would

Box 82.4 Institute of Medicine Recommendations With Regard to Increasing Postmortem Organ Donations

Recommended Against the Following Approaches

- Provision of financial incentives
- Assignment of priority to persons who registered for post-mortem organ donation if they ever need an organ transplant
- Requirement for mandated choice forcing people to commit with regard to their willingness to be a postmortem organ donor
- Presumption of consent to postmortem organ donation with an opt-out provision

Recommended in Favor of the Following Approaches

- Enhanced public education about organ donation
- Influencing of the sociocultural atmosphere to support an expectation of donation
- Simplification of ways for people to register to be organ donors
- Expansion of the efforts of state donor registries
- Improvement in organ procurement organizations' systems so that best practices result in a higher percentage of families' giving consent for organ donation

Data from Childress JF, Liverman CT, eds. *Organ Donation: Opportunities for Action*. Washington, DC: National Academies Press; 2006.

likely still be sufficient for their lifetime, thereby optimizing overall kidney years without sacrificing aggregate recipient life-years.⁶⁰

Although they were not intended or explicit, troubling disparities have resulted from the transplant allocation system. Studies have found that blacks are less likely than whites to be deemed appropriate candidates for transplantation, to be referred for transplant evaluation, to complete a transplant evaluation, to be added to a transplant waiting list, and to receive a kidney transplant.⁶¹ Socioeconomically disadvantaged patients may be delayed in applying for the waiting list, so spend many more years on dialysis. In an effort to reduce these inequalities, the system continues to adjust its allocation criteria. The New Kidney Allocation system amended the definition of wait time to start at the time of listing or retroactively starting at the time of dialysis initiation.⁶⁰ Initial data show that under the new Kidney Allocation system African Americans and Hispanics are receiving a more equitable proportion of transplants.^{60,62}

Transplant of a kidney from a living donor leads to better patient and graft survival than transplant from a deceased donor. However, living-organ donation presents a unique ethical dilemma in that a surgeon must risk the safety of a healthy person to improve the life of a patient. There are three categories of organ donation by living persons: 1. directed donation to a loved one or a friend, 2. nondirected donation to a general pool of patients on a waiting list, and 3. directed donation to a stranger. Each strategy raises ethical concerns. In directed donation, there is the concern that a family member or friend may feel pressured or coerced into donation. Although sometimes admirably altruistic, there is often concern that nondirected donation may be triggered by ethically fraught motivations such as seeking to compensate

for depression, hoping for media attention, or the anticipation of being a part of the recipient's life. There will always be concern about compensation – financial or otherwise – for donation to strangers, especially as there is no national organization to regulate it.⁶³ Although the United Network for Organ Sharing bars directed donation based on race, sex, religion, or national origin, there are many examples of infringements on these directives.^{63,64}

In an attempt to address some of these concerns, a statement from the Live Organ Donor Consensus Group includes requirements for the circumstances of live donation. The donor must be competent, willing to donate, free from coercion, and medically and psychosocially suitable. He or she must also be fully informed of the risks and benefits of being a donor, as well as the risks, benefits, and alternative treatment options for the recipient. Further, the benefits to both the donor and the recipient must outweigh the risks associated with the donation and transplantation.⁶⁵ In 2006, an Institute of Medicine committee cited the need for scientific evidence about long-term risks to donors to help inform this consent process.⁶⁶

The economic pressure inherent in a limited resource of high value caused patients to seek new markets abroad. This created a “transplant tourism” culture in which wealthy patients purchase organs from vulnerable populations of lower-resourced countries. The Declaration of Istanbul on Organ Trafficking and Transplant Tourism strictly prohibited the sale of organs as well as travel to a country for the purpose of obtaining an organ citing violation of the principles of justice and respect for human dignity.⁶⁷ With the notable exception of Iran, organ sale is now illegal across the world. Despite this prohibition, black markets flourish. Poor, indebted donors are subjected to dangerous and unregulated surgeries with frequent infectious complications and long-term poor health. They often receive less compensation than advertised, which is rarely sufficient to create lasting changes in their economic situation. Studies on recipient outcomes are more equivocal, but some have shown more severe infection and more frequent hospitalization.⁶⁸ In 1988, Iran adopted a regulated living-unrelated donor program with a resulting rapid increase in transplants and elimination of the waiting list by 1999.⁶⁹ The system addressed some ethical issues by replacing a black market with regulated practice and by the provision of health care for donors. Charitable contributions limit the disparity between rich and poor recipients. However, donations continue to be from a vulnerable lower socioeconomic group and the financial incentive is still rarely translated into lifelong benefit.

Renal transplantation remains a field rife with active ethical challenge. With unsettled questions of how to best ration limited but life-saving organ resources, and the uncertain territory of a physician's moral stance toward subjecting a healthy donor to a surgical procedure, it is likely to continue to drive ethical debate.

THE “DIFFICULT” DIALYSIS PATIENT AND CONFLICT MANAGEMENT

In recent years, dialysis units have been treating an increasing number of disruptive patients. These patients may be verbally or physically abusive, nonadherent to medical advice, or have substance abuse problems. Their behaviors may interfere

Box 82.5 Strategies for Dealing With the Disruptive or Difficult Dialysis Patient⁷³**Strategies for Working With the Patient**

1. Learn the patient's story and seek to understand his or her perspective.
2. Identify the patient's goal for treatment.
3. Share control and responsibility for treatment with the patient.
 - Educate the patient so that he or she can make informed decisions.
 - Involve the patient in the treatment as much as possible.
 - Negotiate a behavioral contract.
4. Appoint a patient advocate.

Strategies for Preparing the Staff

1. Teach the staff not to criticize patients and call them names.
2. Have the staff use "reflective listening" to show that the patient has been heard.
3. Deal directly with problem behavior; involve the patient, build on the patient's strength, and be clear about who is to do what and when.
4. Take a nonjudgmental approach.
5. Focus on the issue that started the disagreement.
6. Detail the consequences of aberrant behavior in terms that are comprehensible.

7. Prepare a behavior contract that specifies what is to be done by the patient and renal team.
8. Prepare in advance to manage anger.
9. Be patient and persistent.
10. Do not tolerate verbal abuse.
11. Outline for the staff step by step means of coping with agitated and disruptive patients.
12. Establish and publicize a grievance procedure.
13. After effective resolution of a conflict, follow up with the patient to monitor the patient's progress and demonstrate to the patient the commitment to resolve a conflict.
14. Contact law enforcement officials when physical abuse is threatened or occurs.
15. As a last resort consider transfer of the patient to another facility or discharge.
16. Consult with legal counsel before proceeding with plans for discharge and do not discharge the patient without advance notice and disclosure of future treatment options.
17. Contact the end-stage renal disease network if satisfactory resolution has not occurred with the use of these strategies.

From Johnson CC, Moss AH, Clarke SA, Armistead NC. Working with noncompliant and abusive dialysis patients: Practical strategies based on ethics and the law. *Adv Ren Replace Ther.* 1996;3:77–86.

with their own treatment, affect the welfare of other patients and staff, or disrupt the smooth running of the center.^{70,71} The medical community acknowledged the strong ethical and legal obligations toward the patient who requires life-sustaining dialysis treatment. At the same time, the ethical principles of respect for patient autonomy, beneficence, nonmaleficence, and justice command an equal moral weight with respect to other patients and staff on the unit. Dialysis center caregivers report that they feel undertrained and that they lack guidelines to address these situations, causing moral distress and fatigue.⁷⁰ As early as 1990, the "difficult" dialysis patient was identified as one of the top three ethical challenges facing nephrologists.⁷² In 1998 when the Renal Physicians Association and the American Society of Nephrology were developing their Shared Decision-Making clinical practice guideline, the Centers for Medicare and Medicaid Services (CMS) advocated for an entire chapter devoted to such patients (Box 82.5).⁵⁰

The first step in approaching a clash between obligations toward a difficult or disruptive dialysis patient and other values of a dialysis center is to try to change the situation to reduce ethical conflict. The ESRD community collaborated in the Decreasing Dialysis Patient–Provider Conflict project to develop resources promoting understanding, education, and ability to cope with patient–provider conflict in an effort to improve staff–patient relationships and create safer dialysis facilities. The goal was to increase awareness and improve skills to ameliorate conflict including creating a common language to describe and classify conflicts.⁷¹

At times, despite efforts to reduce conflict, the ethical pull to respect a patient's preferences and health continues to compete with the duty to protect welfare and rights of other patients and staff. Nephrologists and other dialysis personnel must use their judgment to balance their moral

obligations as they affect multiple stakeholders. A spectrum of disruptive or difficult dialysis patients ranges from the patient whose behavior harms only himself to one whose behavior endangers other patients and staff in the dialysis unit.⁷³ Ethical obligation will vary depending on where a patient's behavior falls on this spectrum. Mere nonadherence should not usually lead to denial of treatment or discharge from the dialysis unit. Even in cases involving verbally or physically abusive patients, discharge from a dialysis unit should be undertaken only as a last resort after the other strategies presented in Box 82.5 have been exhausted.⁷⁴ This approach allows health-care professionals to identify the limited situations in which involuntary patient discharge from a dialysis unit is ethically justified.⁷⁴

Clinical Relevance

Dialysis units have been treating an increasing number of disruptive and difficult-to-treat patients. Even in cases involving verbally or physically abusive patients, discharge from a dialysis unit should be undertaken only as a last resort.

FUTURE**FINANCIAL STRESSES AND COST-CONTAINMENT STRATEGIES IN NEPHROLOGY**

At the time of its implementation in the 1970s, the ESRD program was expected to incur modest costs compared with the value of a life-saving treatment. However, early cost expectations were quickly outpaced by an expanding

dialysis population, and optimism for the treatment effects was tempered by increasingly apparent limitations on the quality of life that could be restored to ESRD patients on dialysis. Since the establishment of the Medicare ESRD program, high costs throughout the entirety of Medicare became a more pressing concern. In the context of rising health-care costs in the late 1980s in the United States, critics wondered if more good could be done by investing resources in other chronic disease programs.⁷⁵ They raised concern that the ESRD program was not sufficiently fair and was not the best use of limited financial resources. In other words, it satisfied neither of the first principles of distributive justice: equality and utility. First, it was unclear why governmental support should privilege one group of patients with chronic illness—those with ESRD—while there was no comparable program for failure of the heart, lungs, or liver.^{75,76} It was difficult to justify the use of 7.2% of the total 2013 Medicare budget for <1% of the Medicare population while millions of Americans go without basic health-care insurance.²⁹ From a cost-effectiveness perspective, mounting data showing that dialysis was not the rehabilitating therapy that it had been hoped was discouraging. For many, life on dialysis was synonymous with physical incapacitation, dependency, chronic depression, and disrupted family functioning.⁷⁷ Further, the 3-year adjusted survival rate for an HD patient in 2011 had increased slowly to only 59%.²⁹

Short of dismantling the program, the Centers for Medicare and Medicaid proposed significant payment cuts to dialysis facilities in 2013 and a variety of strategies for cost containment have been subsequently suggested.⁷⁸ The Affordable Care Act established ESRD Seamless Care Organizations (ESCOs) giving dialysis providers and centers an incentive to promote value and cost containment by their sharing in the financial benefits of savings and burdens of spending.⁷⁹ (The ethical challenges of ESCOs are discussed in more detail later.) Others have pointed to the extreme waste in the medical system as a focus for future cost reduction.

Given high potential savings, the prospect of rationing has also emerged in discussions of cost containment. Age-based rationing has been suggested because elderly patients are a growing portion of the dialysis population, their care is more expensive, and they have a shortened survival.⁸⁰ However, rationing by age is prohibited by the Age Discrimination Act and regardless is not thought justifiable by most ethicists.⁸¹ Covert age rationing of dialysis by the British National Health Service raised the ire of the British public, with resultant regulations to improve dialysis services provided to the elderly to be more consistent with neighboring European countries.⁸² While overt rationing seems unlikely to gain public or political traction, an increased cost awareness necessitates creative thinking on how to integrate financial considerations into health care without sacrificing quality. In the United States, Medicare's National Quality Strategy advocates a balance of multiple values by endorsing three aims: 1. better care for the individual, 2. better health for populations, and 3. reduced health-care costs.⁸³ An example of such a strategy includes starting dialysis later in the disease course, which has been shown to have similar patient outcomes and to be more cost-effective. Others have advocated greater adoption and support for peritoneal dialysis (PD) given that it has been shown to have comparable rates of survival and complications, and is less expensive compared with HD.⁸⁴

ESCOS AND CHERRY PICKING OF DIALYSIS PATIENTS

In a new effort to improve quality and control the cost of the Medicare ESRD Program, the CMS have implemented the Comprehensive ESRD Care Model.⁸⁵ CMS is partnering with large dialysis organizations to test the effectiveness of a new payment and service delivery model in providing beneficiaries with person-centered, high-quality care. This ESCO model is built on the Accountable Care Organization (ACO) experience from the Pioneer ACO Model, Next Generation ACO Model, and the Medicare Shared Savings Program. However, while the attempt to improve quality of care and respect for cost is laudable, potential conflicts of interest arise. Under the existing system of value-based purchasing, in which dialysis centers are held accountable for their achievements on quality metrics, there has been a temptation to enrich one's pool with relatively healthy dialysis patients. Because ESCOs are financed based on a shared risk model, large dialysis organizations have an even stronger incentive to cherry pick healthy patients as a large percentage of their income is at stake. Older and sicker dialysis patients require more dialysis center staff time and resources and are more frequently hospitalized. Both detract from a dialysis organization's income in a shared risk model. Under the current program, patients already established with a dialysis unit cannot be excluded when it becomes an ESCO, but nephrologists and unit administrators may be tempted to encourage older, sicker, and therefore more expensive patients to withdraw from dialysis. Commentators have called for "ethically transparent" approaches for deciding who does and does not start dialysis and for meaningful shared decision-making based on evidence-based guidelines to resist the temptation to cherry pick.⁸⁶

PROFESSIONAL ETHICS IN DIALYSIS

A significant portion of many nephrologists' income comes from practice in dialysis centers and often a share in ownership. Ownership offers the opportunity for a nephrologist to direct the mission of a center toward patient goals, but also brings conflict of interest and potential breach of professional ethics. Decades ago, nephrologist and then *New England Journal of Medicine* editor Arnold Relman anticipated this predicament. He warned that the private enterprise system—the so-called new medical-industrial complex—had a particularly striking effect on the practice of dialysis, and he urged physicians to separate themselves totally from any financial participation so as to maintain professional integrity.⁸⁷

As dialysis centers expand and the market becomes saturated, there is intense pressure to recruit patients. A nephrologist who encounters a patient during a hospital admission might encourage the patient to transfer to a different dialysis center in which he has a financial interest. She may suggest that her own dialysis center is better when there are no objective quality measures to support the assertion. This practice of patient solicitation disrespects the physician-patient relationship, informed consent, continuity of care, and is an example of clear conflict of interest. It has been roundly criticized as falling short of professional ethical conduct.^{88,89}

Dr. Relman's strict entreaty to completely separate practice from financial interests may not be entirely achievable because both nephrologists and dialysis centers do need to maintain

healthy financing to continue to provide care. However, medicine as a professional practice is held to a different ethical standard than behavior in other marketplaces. As compared with businessmen engaging in a free-market enterprise, professionals have a fiduciary duty to prioritize the well-being of the customer-patient. With this in mind, nephrology societies have established recommendations around professional ethics in dialysis practice. Nephrologists have a duty to give patients evidence-based information about the quality of dialysis centers and treatment options as well as fully disclose their personal interests which might interfere with professional judgment.⁸⁸

A GLOBAL PERSPECTIVE ON NEPHROLOGY HEALTH CARE

Patterns and norms directing dialysis use in the United States are all the more stark in contrast to other developed countries. For example, in comparing practices between a population in the United States and Australia, rates of treatment in patients aged under 45 years were similar, but rates of treatment for patients greater than 85 years of age were 41% in the United States compared with 6.8% in Australia.³⁵ Japan, Taiwan, and Singapore also manifest a pattern of higher rates of dialysis in the elderly ESRD population, whereas other comparably high-income countries have lower prevalence.⁹⁰ These differences emphasize the context and value-dependent nature of what is considered ethical policy and practice.

As life expectancy increases through social and medical advances worldwide, chronic diseases including diabetes and hypertension become more prevalent. The rate of ESRD has similarly risen sharply. One systematic review estimates that in 2010, 2.6 million people received dialysis worldwide, but that nearly the same number died without access to the treatment.⁹⁰ They further anticipate a doubling of the global prevalence of ESRD by 2030, with growth largely in regions with limited access. KDIGO guidelines on supportive care draw attention to low-income populations as a distinct group with “choice restriction,” emphasizing the importance of developing policy aligned with the realities of differing resources in high- and low-income countries.⁴⁵ One strategy of cost containment while maintaining quality is the “PD first initiative” exemplified by Hong Kong. Patients are informed and free to choose either PD or HD, but only PD will be reimbursed if there is no contraindication to this therapy. This practice makes Hong Kong one of the only countries to have a greater proportion of PD than HD patients.⁹¹ Given the similar quality of life and efficacy of PD there are many reasons the modality is better suited to resource-limited settings. It is simpler, requires fewer trained staff, does not require electricity or large volumes of filtered water, and is importantly potentially cost saving in comparison with HD.⁹¹ As individual nations develop a more connected global community, disparities between resource-rich and resource-poor countries are more apparent, and the question of the ethical responsibilities of a resource-rich country to a resource-poor country becomes more poignant.

ETHICAL CHALLENGES POSED BY GENETIC ADVANCES IN CKD

Better understanding of the genetic factors that predispose to particular diseases has revolutionized detection and

counseling for high-penetrance genetic disorders such as autosomal dominant polycystic kidney disease. However, with this information come complex questions about how to convey risk information to presymptomatic carriers, the duty to warn genetically related family members, and how this information should inform reproductive choices. In 1997, the World Health Organization (WHO) proposed international guidelines on many ethical issues in medical genetics. These include protocols for informed consent for testing, disclosure and confidentiality, and presymptomatic testing.⁹² They highlight the unique aspect of genetic testing which involves people beyond the patient. For example, family members may not have personally sought out medical information, but their lives might be changed by it nonetheless. Unfortunately, recent U.S. history gives examples of mishandling of genetic information that led to social stigma and mistrust of the healthcare community. In the 1970s, sickle cell screening was implemented before a clear understanding of any clinical significance of carrier status, and it caused stigmatization of many patients of African descent, seeding distrust of medical providers within this community. The WHO emphasizes the imperative to never use genetic data to stigmatize or discriminate.⁹²

Genome-wide sequencing technology has the potential to identify new genetic risk factors for kidney disease. As an example, certain APOL1 alleles have been associated with a 1.8 hazard ratio of progressive kidney disease among the African American population.⁹³ While the prospect of genetic tests moving from laboratory bench to clinical practice is exciting, it also raises ethical concerns. For example, should genetic testing be performed if the prognostic significance and/or therapeutic options have yet to be clarified? Potential genetic risk factors that are more prevalent in particular populations such as APOL1 in those of African descent risk causing health-care disparity. For example, APOL1 has been proposed as a screening tool for transplant donor candidates to protect those who are at higher risk for future kidney disease. However, because the risk allele is found exclusively in those of African descent, a selected exclusion of many black donor candidates could cause disparity in black donors' ability to offer organs to friends and family. Further, as the medical utility of the concept of race comes into question, it will be ethically challenging to design population-level uses for genes that are shown to cosegregate by distant genetic heritage. Anticipating that new advances will bring novel ethical dilemmas, the WHO-proposed guidelines emphasize the importance of interval modification over the years.⁹² Collectively, these concerns remind us that medicine is not practiced by healthcare providers in ethical isolation, and it continues to be important to incorporate diverse community perspectives in defining research priorities, policy development, and clinical practice.

CONCLUSION

The ethical issues facing nephrologists in the 1960s ushered in the field of bioethics. Despite being decades old, the ESRD program in the United States is still confronted with new ethical challenges including uncertainty about what is a fair proportion of federal funding for the Medicare ESRD program to receive, how to counter unintended incentives

to cherry pick dialysis patients amid new funding structures, what are appropriate responses to difficult patients, and how to care for an increasingly elderly and multimorbid population at the end of life. With the growth in the CKD and ESRD population, the changes in its demographics, and the economic challenges to the Medicare ESRD program posed by a large federal government budget deficit, sustainable and equitable financing strategies are needed. In transplantation, the intricate considerations related to putting a living kidney donor's health at risk for the benefit of a kidney transplant recipient continue to be ethically complex. As the demand for transplantation donors continues to rise and as waiting times on the deceased donor transplant list lengthen, new strategies to increase the number of donors while being careful to regulate equitable distribution stretch our ethical foundation. Given new challenges posed by scientific discoveries, there is every expectation that nephrologists will continue to be at the forefront among those facing ethical challenges in medicine.

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BOARD REVIEW QUESTIONS

1. Which one of the following statements best summarizes the role of shared decision-making for patients with chronic kidney disease (CKD)?
 - a. Shared decision-making is an outmoded concept from the 1980s.
 - b. Shared decision-making fits well with a disease-oriented approach to CKD patient treatment.
 - c. Shared decision-making for CKD patients defaults to dialysis modality choices.
 - d. Shared decision-making is the recognized preferred model for medical decision-making.

Answer: d

Rationale: Although shared decision-making was introduced in the 1980s, it currently remains the recognized preferred model for medical decision-making. It was recommended in 2010 in the second edition of the Renal Physicians Association clinical practice guideline and the American Society of Nephrology Choosing Wisely Campaign in 2012. Shared decision-making has been called the “pinnacle of patient-centered” care. It does not fit well with a disease-oriented approach. It concerns much more than dialysis modality.

2. Which one of the following is true with regard to informed consent for dialysis for an older stage 5 chronic kidney disease (CKD) patient with significant comorbid conditions?
 - a. An “age-neutral” approach is appropriate to avoid discrimination.
 - b. It should include disclosure that dialysis may not confer a survival advantage.
 - c. Consent for dialysis is a formality because stage 5 CKD patients want to live.

- d. Physicians are to provide information, not to make a recommendation.

Answer: b

Rationale: Multiple studies have found that for older patients with significant comorbidities dialysis may not confer a survival advantage. Because the research findings particularly address CKD patients older than 75, an “age-neutral” approach has been discouraged. Because of the additional considerations with regard to quality of life and survival for older patients, a detailed informed consent discussion is necessary. It is not merely a formality. The ethical model of informed consent requires physicians to make a recommendation, not just provide information.

3. Which one of the following ethical principles justifies patient self-determination with regard to deciding to initiate or forgo dialysis?
 - a. Beneficence
 - b. Justice
 - c. Nonmaleficence
 - d. Respect for autonomy

Answer: d

Rationale: Respect for patient autonomy is the ethical principle which is based on the concept that people should be empowered to make decisions for themselves in an uncoerced manner to the extent that they are able to understand their health condition and treatment options with their attendant risks and benefits. The legal right of patient self-determination is based on this principle. Beneficence, justice, and nonmaleficence refer to ethical principles to promote patient benefit, treat patients fairly, and do no harm, respectively.